

- **Alibaba:** Druid is used for performing real-time analytics of users' interactions on Alibaba's e-commerce website.
- **British Telecom:** Druid was implemented at British Telecom after conducting a detailed comparison across various parameters such as ingestion, read performance, aggregation, high availability, etc. A detailed report on this is available at <https://imply.io/videos/why-british-telecom-chose-druid-over-cassandra>.
- **Cisco:** At Cisco, Druid powers the network flow data analytics in real time.
- **Netflix:** Druid is used to aggregate the many data streams at Netflix. A detailed article is available at <https://netflixtechblog.com/how-netflix-uses-druid-for-real-time-insights-to-ensure-a-high-quality-experience-19e1e8568d06>, which explains how Druid is being used by Netflix for providing real-time insights to enable a high-quality experience for its customers.
- **PayPal:** Druid has been adapted at PayPal to handle massive volumes of data. This is used for internal exploratory analytics.
- **Walmart:** Druid is employed at Walmart for event stream analytics. Further details are available at <https://medium.com/walmartglobaltech/event-stream>

analytics-at-walmart-with-druid-dcf1a37ceda7.

Ingestion

The term ingestion has been used a few times in this article. Before the actual installation of Druid, let's understand what ingestion is and what its types are.

Druid organises data in the form of segments. Each of these segments is a data file. These files carry a few million rows. Loading of data in Druid is termed ingestion, and consists of the following:

- Reading data from the source system
- Building of segments based on this data

Ingestion can be carried out in two ways:

- Streaming ingestion
- Batch ingestion

Each of these ingestion systems has support for its own sources from which it pulls the data. Streaming ingestion is recommended and popular. It reads data directly from Kafka. The Kinesis indexing service is also used.

In the case of batch indexing, there are three options:

- *index_parallel*
- *index_hadoop*
- *index*

To understand Druid better, it is important to have an idea about its design and components. With a primarily multi-process and distributed

architecture, Druid is cloud-friendly in its design.

Druid processes and servers

There are many types of processes as listed below.

- **Coordinator:** Manages data availability on the clusters
 - **Overlord:** For controlling the assignment of data ingestion workloads
 - **Broker:** This is for handling queries from external clients
 - **Router:** Used to route requests to brokers, coordinators and overlords
 - **Historical:** For storing the data that can be queried
 - **Middle manager:** This is responsible for ingesting data
- There are three types of servers.
- **Master:** This is for running the coordinator and overlord processes. It also manages data availability and ingestion.
 - **Query:** This runs broker and router processes (optional). It is used for handling queries from external clients.
 - **Data:** This runs the historical and middle manager processes. It is used to execute ingestion workloads.

External dependencies

Druid has three external dependencies.

- **Deep storage:** This is a shared file storage which is accessible by each Druid server. This can be a distributed object store such as S3 or a network mounted file system.
- **Metadata storage:** This is used for storing various shared system metadata. This can include items such as segment usage information and task information.
- **ZooKeeper:** This is used for various activities such as internal service discovery, coordination and leader election.

The architecture diagram of Druid (from the official documentation) illustrates the components and their interaction (Figure 4).

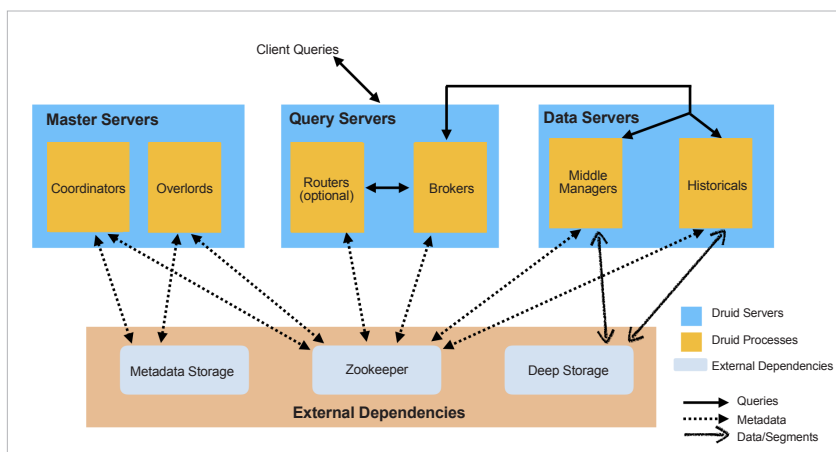


Figure 4: Architecture of Druid